Class 5–6: Operating System, Software & Computer Security

1. Introduction to Computer System

A computer system works with the help of two main components: **Hardware** and **Software**.

- **Hardware** is the physical part (keyboard, CPU, monitor).
- **Software** is the set of instructions that tells the hardware what to do.

A computer cannot function properly without software, and the most important software is the **Operating System (OS)**.

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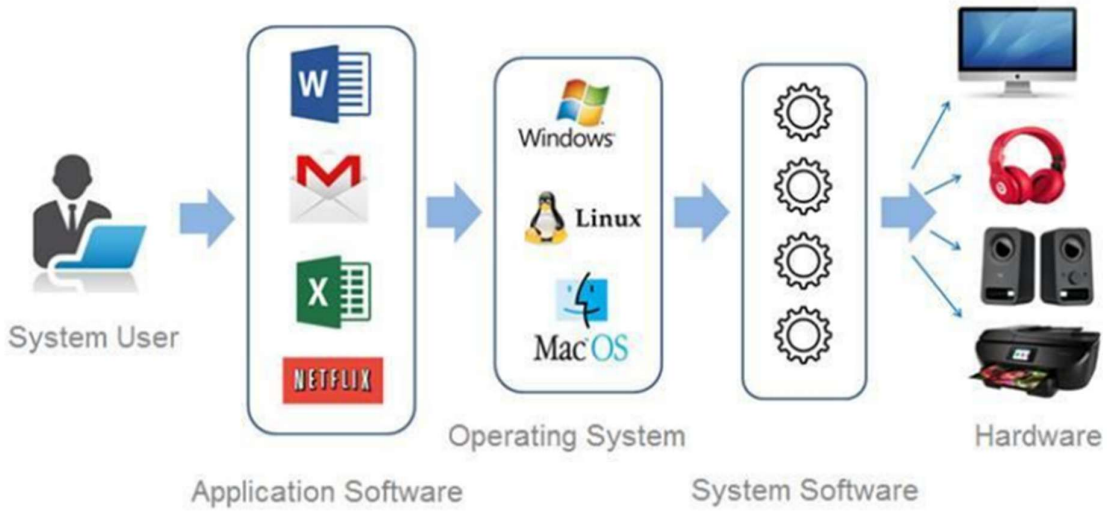
2. What is an Operating System (OS)?

Definition:

An Operating System (OS) is a system software that acts as a **bridge between the user and the computer hardware**.



Why OS is Important:

- It makes the computer usable
- It controls all hardware components
- It allows users to run applications
- Without OS, computer is useless

Basic Working of OS:

- User gives input
- OS processes the requestOS instructs hardware
- Output is shown to user

Real-Life Example:

Think of the OS as a **manager**:

- User = Boss
- OS = Manager
- Hardware = Workers

The manager takes instructions from the boss and tells workers what to do.

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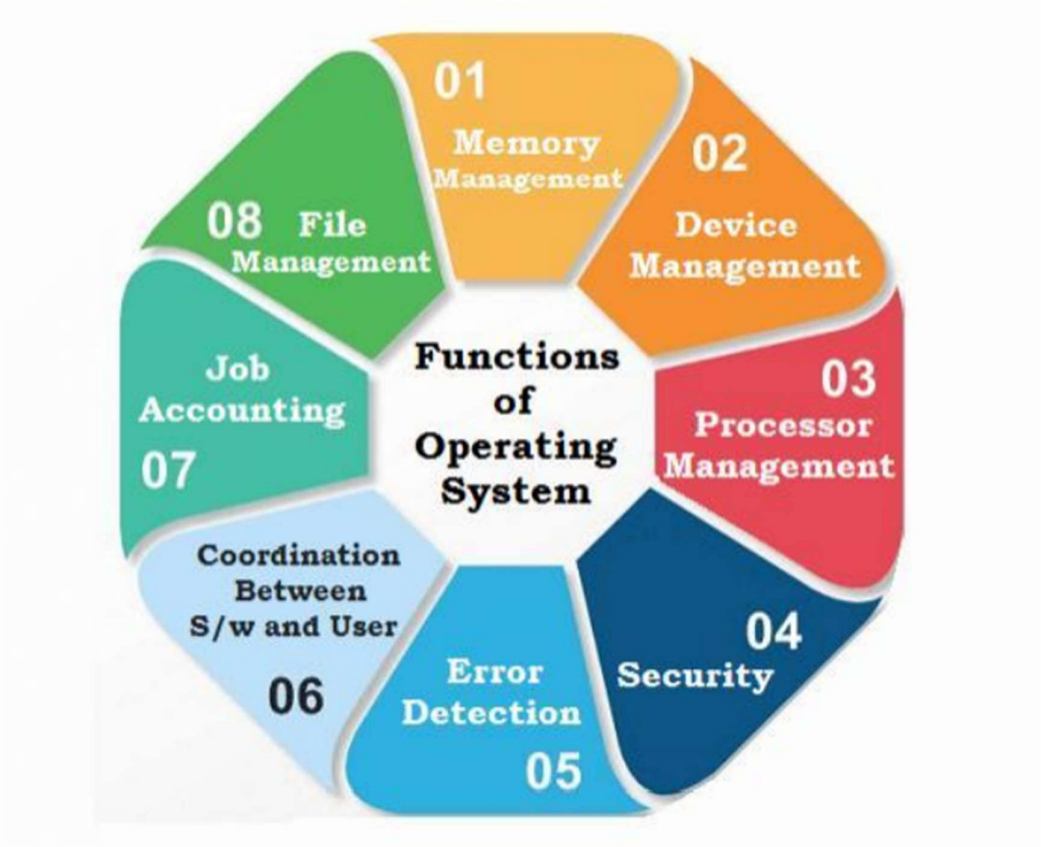
3. Functions of Operating System



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1. Process Management

- Runs and manages programs
- Allows multitasking (multiple apps at same time)
- Decides which program runs first

Example: Listening to music while using browser

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2. Memory Management

- Manages RAM (temporary memory)
- Allocates memory to different programs
- Prevents memory conflicts

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3. File Management

- Organizes files and folders
- Helps in storing and retrieving data

Example: Saving files in folders like Documents

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4. Device Management

- Controls input and output devices
- Uses drivers to communicate with hardware

Examples: Keyboard, mouse, printer

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5. Security Management

- Protects system using passwords and permissions
- Controls access to files and programs

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4. Types of Operating Systems

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1. Single User OS

- Used by one user at a time
- Common in personal computers

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2. Multi User OS

- Multiple users can use system simultaneously
- Used in servers and organizations

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3. Multitasking OS

- Runs multiple applications at once
- Improves efficiency

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4. Real-Time OS (RTOS)

- Provides instant response
- Used in critical systems

Examples: Robotics, medical systems

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5. Common Operating Systems

- **Windows:** Used in PCs and laptops
- **Android:** Used in mobile phones
- **macOS:** Used in Apple devices
- **Linux:** Open-source, used in servers and advanced systems

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6. User Interface (UI)

User Interface is the way users interact with the computer.

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1. GUI (Graphical User Interface)

- Uses icons, buttons, windows
- Easy to use

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2. CLI (Command Line Interface)

- Uses text commands
- More powerful but difficult

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7. Booting Process

What is Booting?

Booting is the process of **starting the computer and loading the OS into memory (RAM)**.

Steps of Booting:

1. Power button is pressed
2. BIOS starts and checks hardware
3. OS loads into RAM
4. System becomes ready to use

What is BIOS?

- Basic Input Output System
- It checks hardware before OS starts

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8. Proper Shutdown

What is Shutdown?

Shutdown is the process of safely turning off the computer.

Why Proper Shutdown is Important:

- Saves data
- Closes programs properly
- Prevents system damage

Risks of Improper Shutdown:

Data loss

- File corruption
- Operating System damage

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9. Understanding Software

Definition:

Software is a set of instructions that tells the computer what to do.

Concept:

- Hardware = Body
- Software = Brain

Role of Software:

- Makes hardware useful
- Helps perform tasks
- Controls operations

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10. Types of Software

1. System Software

- Controls and manages computer
- Runs in background

Examples: Operating System, Device Drivers

2. Application Software

- Used by users for specific tasks

Examples:

- Web browser (Chrome)
- Word processor (MS Word)
- Media player (VLC)

Difference Between System & Application Software

Feature	System Software	Application Software
Purpose	Runs computer	Performs user tasks
Usage	Background	On demand
Example	OS	Chrome, Word

11. Computer Virus & Security

What is a Virus?

A computer virus is a harmful program that damages the system or steals data.

Effects of Virus:

- Deletes files
- Slows system
- Shows unwanted ads
- Steals information

How Viruses Spread:

- Downloading from unknown websites
- Using infected pen drives
- Clicking unknown links
- Installing pirated software

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12. Protecting Your Computer

Antivirus Software

Software that detects and removes viruses

Examples: Windows Defender, Quick Heal

Safety Practices:

- Scan pen drives before use
- Keep system updated
- Avoid unknown downloads
- Do not click suspicious links

Golden Rule:

"Free software can be costly"

(It may contain hidden viruses)

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13. AI and Computer System

Connection with AI:

- AI tools are also software
- They run on OS
- Require good system performance

Important Points:

- High RAM improves AI performance
- SSD increases speed
- Secure system ensures safe AI usage

Assignment 1: Operating System Explorer

Teach & Check Activity

Person You Explained To: _____

1. Which interface did they prefer? (GUI / CUI)

2. Why did they find it easy or difficult?

Reflection

1. What confusion did they have?

2. How did you explain it?

3. What did you learn while teaching?

Assignment 2: Inside Your Computer – BIOS & Startup Mystery

Understanding Concepts

1. What is BIOS (in your own words)?

2. What happens when you press the power button? (Step-by-step)

1. 2. 3. 4.

3. What is Booting Process?

Real-Life Scenario Task

If computer is switched off directly from power button daily:

Problems that can happen:

> -

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Why does this happen?

Teach & Check Activity

Explain: “Why proper shutdown is important”

1. Did they already know this?

2. What new thing did they learn from you?

Assignment 3: Operating System Comparison Challenge

Comparison Table

Feature	Windows	Android	iOS
Device Type	_____	_____	_____
Price	_____	_____	_____
Speed	_____	_____	_____
Ease of Use	_____	_____	_____
Customization	_____	_____	_____
Security	_____	_____	_____

Additional Details

Windows

- Latest Version: _____
- Advantages: 1. 2.
- Disadvantages: 1. 2.
- Preferred Users: _____

Android

- Latest Version: _____
- Advantages: 1. 2.
- Disadvantages: 1. 2.
- Preferred Users: _____

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iOS

- Latest Version: _____
- Advantages: 1. 2.
- Disadvantages: 1. 2.
- Preferred Users: _____

Analysis Questions

1. Which OS is best for students and why?

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2. Which OS is most secure and why?

3. Which OS gives more control/customization?

4. If budget is low, which OS is better and why?

Teach & Check Activity

Explain any one OS to someone

1. What questions did they ask?

2. What confusion did they have?

3. How did you explain it?

Smart Decision Task

1. Which OS will you choose and why?

2. What features are most important for you?

3. Which OS will you avoid and why?

Final Section: My Learning Advantage

3 Things I Can Explain Confidently:

1 Thing I Want to Improve: